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Meta-analysis

Sugar-sweetened beverages consumption increases the risk of metabolic syndrome and its components in adults: Consistent and robust evidence from an umbrella review

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SUMMARY

Background & aims: Sugar-sweetened beverages (SSBs) are potential modifiable risk factors for reducing the risk of metabolic syndrome (MetS); however, there is a lack of an umbrella review that robustly synthesizes the evidence. This review aims to synthesize the evidence linking habitual SSBs intake with MetS in adults, emphasizing the need for policy and regulatory actions.

Methods: Four databases were comprehensively searched for relevant meta-analyses of prospective studies up to July 2023. Two authors independently conducted the screening for eligibility, data extraction, and quality assessment. The outcome of interest was MetS in adults, encompassing its component conditions. The present review was registered with PROSPERO (CRD42023402549).

Results: Of the 16 eligible meta-analyses identified, 30 summary estimates were obtained for the impact of SSBs consumption on obesity, type 2 diabetes, hypertension, and MetS. Seven were rated as 'High' methodological quality, with the rest classified as 'Moderate' and 'Low' quality, consisting of three and six references, respectively. A comparison of the highest and lowest levels of SSBs consumption revealed an increased risk of 18% (95% CI: 13%–24%), 12% (95% CI: 11%–14%), 29% (95% CI: 25%–32%), and 29% (95% CI: 7%–52%) for obesity, hypertension, type 2 diabetes, and MetS, respectively. Consistently, the findings from dose–response analyses are in agreement with and corroborate the existing evidence that SSBs are a significant risk factor for the development of MetS and its related conditions. Noticeably, the quality of evidence was predominantly deemed as highly suggestive and convincing. Moreover, consensus on specific criteria to identify studies related to SSBs in literature searching was lacking, and most primary studies were conducted in developed countries and Europe.

Conclusions: Our findings suggest that more rigorous and targeted policy interventions are warranted to curtail SSBs consumption, in order to alleviate the global burden of MetS.

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Abbreviations: BMI, Body mass index; CIs, Confidence intervals; HFCS, High-fructose corn syrup; LMICs, Low- and middle-income countries; MeSH, Medical Subject Headings; MetS, Metabolic syndrome; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; SSBs, Sugar-sweetened beverages; USA, United States of America; WHO, World Health Organization; WC, Waist circumference.

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1. Introduction

Non-communicable diseases (NCDs) have had a significant impact on health systems and economies worldwide, contributing to increased morbidity and mortality, reduced quality of life, and escalating healthcare expenditures, particularly in low- and middle-income countries (LMICs) [1].

The emergence of metabolic syndrome (MetS) incidence is placing an additional burden on global public health systems, further compounding the already substantial impact of NCDs on health systems and economies worldwide [2]. While there may be some minor variations in the definition of MetS by different healthcare organizations, it is generally recognized as a cluster of comorbid conditions, including obesity, hypertension, and disorders of carbohydrate and hyperlipidemia [3]. The global prevalence of MetS exhibits considerable regional variation, partly attributable to differences in lifestyle behaviors and ethnicities [4]. According to estimates, the prevalence of MetS ranges from 20% to 25% globally [5]. Regionally, the prevalence of MetS among the Asian population ranges from 12% to 37%, while in the European population, it ranges from 12% to 26% [6]. Of note, the compositions of MetS reveals its significant impact on adult health worldwide, with more than 1.9 billion adults being overweight, over 650 million classified as obese (in 2016) [7], around 1.28 billion affected by hypertension (in 2023) [8], approximately 464 million with impaired glucose tolerance (in 2021), around 298 million with impaired fasting glucose (in 2021) [9], and a prevalence of about 39% for raised total cholesterol among adults (in 2008) [10]. The pathophysiology of MetS is complex and multifactorial, involving an interplay of various factors such as caloric imbalance, genetic and epigenetic determinants, sedentary lifestyles, and dietary composition, as well as the gut microbiome. Among these factors, an imbalance of calorie intake and energy expenditure is recognized as a significant contributor to the development of MetS, with food and beverage consumption being an essential modifiable factor in its pathogenesis [2]. Noticeably, the World Health Organization (WHO) has identified SSBs consumption as a primary risk factor for the aforementioned chronic diseases and has recommended reducing their intake to promote better health outcomes [11].

The ubiquity and variety of calorie-containing beverages, such as SSBs, have led to a notable rise in their consumption, thereby increasing their contribution to total energy intake among individuals [12]. This rise in SSBs consumption has raised concerns regarding their potential health effects. The prevailing consensus among scholars and public health professionals is that the deleterious health outcomes attributed to SSBs consumption are not confined to the elevated risk of obesity; rather, they also encompass the comorbidities associated with other components of MetS, including type 2 diabetes and hypertension [13]. Therefore, it is essential to comprehensively understand the impact of SSBs consumption on human health outcomes, including the risk for MetS, in order to design effective interventions aimed at its modification. However, it is evident that despite the significant impact of SSBs on human health outcomes and the abundance of data on beverage consumption in various countries worldwide, there has been a notable absence of an umbrella review synthesizing the association between habitual SSBs consumption and MetS among the general adult population.

To address this gap in knowledge, our objective was to systematically and quantitatively synthesize the available meta-analysis evidences on the association between SSBs consumption and MetS in adults. In addition to presenting the meta-analytic results, our review highlights the novel findings where top-tier evidence regarding the impacts of SSBs on adult metabolic health and the current limitations in existing studies in this area are currently unavailable. These results have significant implications for guiding global health practice and policy and identifying research priorities for the future.

2. Methods and materials

The protocol for this review was prospectively registered at the International Prospective Register of Systematic Reviews (Registration number: CRD42023402549). The protocol of this umbrella review was devised in accordance with the guidelines set out by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [14].

2.1. Literature search and selection criteria

Starting from inception up to July 2023, two authors conducted an exhaustive search of four electronic databases, namely PubMed, Scopus, MedRxiv and Epistemonikos to identify relevant articles exploring the plausible associations between the consumption of SSBs and diverse range of health outcomes. The employed terminologies consisted of Medical Subject Headings (MeSH) terms and keywords that were germane to SSBs, meta-analysis, and systematic review. This utilization of a predetermined search strategy is explicated in Supplementary Table S1. In addition, a manual review of the reference lists of the discovered publications was undertaken to uncover other articles that may have relevance to our review. Systematic reviews incorporating meta-analyses of prospective studies exploring the potential associations between SSBs consumption and health outcomes were considered eligible for our inclusion based on the following criteria.

1) The included systematic reviews and meta-analyses comprised of prospective cohort studies that had a minimum follow-up period of one year.
2) The meta-analyses reported the associations between SSBs consumption and MetS conditions in human, particularly in obesity, hypertension, impaired glucose metabolism (prediabetes or diabetes) and atherogenic dyslipidemia [3].
3) Studies including participants aged 18 years or older with no exclusion criteria for health condition, age, gender, ethnicity, or country.
4) The meta-analyses furnished precise data for computing summary effect size, 95% confidence intervals (CIs), and corresponding numbers of cases and population.
5) To enhance the credibility and robustness of the findings, a minimum of two meta-analyses were considered essential for each health outcome variable to be incorporated in our review.

Inclusion in this review was contingent upon studies being written in English. This criterion was adopted given that a majority of well-designed systematic reviews and meta-analyses are published in English.

We implemented a set of criteria to rule out studies, such as: (1) studies involving animal research or in vitro/ex vivo experiments; (2) systematic reviews without a meta-analysis; (3) mini reviews, literature reviews, and umbrella reviews; and (4) studies that were solely protocols, commentaries, abstracts, letters to the editor, or consensus statements. In addition, during the inclusion process, we made post-hoc decisions to exclude studies based on specific criteria, such as: (1) analyses separate type of SSBs (e.g. fruit juices, artificially sweetened beverage, srtificially sweetened soft drinks) rather than consumption of SSBs in general. The rationale behind this approach is to maintain consistency in the research selection criteria and mitigate potential errors or biases related to specific types of SSBs. Consequently, our study exclusively focuses on

general SSBs, encompassing all categories of these beverages, while excluding studies that solely evaluated the effects of individual SSBs on metabolic health (e.g., studies exclusively assessing the health impacts of fruit juices). Additionally, this methodological approach demonstrates practical pragmatism as it considers the realistic scenario where study participants consume various SSBs varieties throughout the study rather than a singular type; (2) systematic reviews that had geographical limitations to studies conducted in a specific country; (3) using alternative units, such as kilograms (kg) or monthly.

The process of selecting eligible articles involved independent screening of titles, abstracts, and full article texts by two authors, with discrepancies being resolved through discussion with a third investigator until a consensus was achieved.

2.2. Data extraction

To collect the required data from each qualifying systematic review, a table with predefined columns was developed. This included key study details such as study identification (first authors and published year), health outcomes, number of studies included, case numbers, total population, age range, sample size range, tracking time, locations, searching keywords, effect size and dose–response analyses with the corresponding 95% CI, effect model (fixed or random), I^2 statistic value, Cochran's Q test P-value, Egger's test P value. If a publication contained distinct meta-analyses for multiple outcomes, each analysis was examined separately and the corresponding outcomes were included as appropriate. In situations where multiple meta-analyses reported the same health outcome among a given population, we priority extracted data from the largest and most recent study. During the data extraction process, three independent researchers conducted the process of data extraction from each eligible systematic review in a blinded fashion. By conversation with a fourth reviewer, we were able to iron out any differences that had been found.

2.3. Grading of the evidence

The meta-analyses were classified into four evidence groups based on the significance of the associations, following the previously outlined criteria [15,16]. Briefly, correlations with nominally significant random-effects summary effect sizes (i.e., $P \leq 0.05$) had their believability rated according to the following categories: convincing (class I), highly suggestive (class II), suggestive (class III), weak evidence (class IV), and insignificant, as shown in detail in Supplementary Table S2. If the specified criteria were not satisfied, the meta-analysis was categorized as indicative of weak evidence.

2.4. Methodological quality appraisal

The methodological quality of the included reviews was evaluated using the AMSTAR 2 (A Measurement Tool to Assess Systematic Reviews) checklist, which enabled an overall rating to be assigned [17]. The AMSTAR 2 questionnaire has sixteen questions to which reviewers are asked to respond "Yes" or "Partial Yes" or "No". The overall quality score for each included study was determined based on the presence of critical and non-critical weaknesses. Within the domains encompassed by AMSTAR-2, seven inquiries are deemed pivotal for appraising the comprehensive quality of the reviews. These inquiries encompass fundamental aspects such as the establishment of an a priori protocol (question 2), adequacy of the literature search (question 4), the rationale for study exclusion (question 7), rigorous evaluation of the risk of bias (question 9), appropriate implementation of meta-analytic methodologies (question 11), incorporation of bias considerations in result interpretation (question 13), and assessment of publication bias (question 15). The included systematic reviews were categorized into four groups based on the severity of their limitations, as follows: high quality (no or only one non-critical weakness); moderate quality (more than one non-critical weakness); low quality (one critical flaw with or without non-critical weaknesses); and critically low quality (more than one critical flaw with or without non-critical weaknesses) [17]. Two independent assessors evaluated the methodological quality of the reviews using the AMSTAR 2 tool, and any differences in scores were resolved through mutual discussion.

2.5. Statistical analyses

For each association between SSBs consumption and health outcomes, we employed the DerSimonian and Laird random-effects model to recalculate the adjusted summary estimates, their 95% confidence intervals, and P values. If dose–response data was available, we recalculated the analyses accordingly. We used the I^2 statistic and the Cochrane's Q test to examine statistical heterogeneity between included studies. An I^2 score over 50% or a P value for the Q-test below 0.1 indicated the existence of substantial between-study heterogeneity. The possibility for publication bias was evaluated using funnel plots and Egger's regression test, with a P-value <0.1 indicating statistical significance [18]. Stata version 16 (Stata Corp, College Station, Texas) was used for all statistical analyses with statistical significance set at a threshold of $P \leq 0.05$. The compilation of search keywords data was performed using MS Office 365 (Supplementary Table S2), and its visual representations were generated using RStudio version 2022.12.0 + 353 with the *"wordcloud"* package.

3. Results

3.1. Study selection and characteristics

Our systematic search strategy yielded a total of 816 unique references, which underwent a rigorous screening process as depicted in Fig. 1. Following an initial screening of titles and abstracts, 715 references were deemed irrelevant and consequently excluded. The remaining 101 references underwent a thorough full-text review for eligibility assessment. We selectively retained the most recent and methodologically rigorous meta-analyses, in accordance with the procedures outlined in the Methods section. Ultimately, we identified 16 publications that satisfied our inclusion criteria and provided significant insights into 30 risk ratio (RR) of meta-analyses investigating MetS and its related conditions (obesity, type 2 diabetes and hypertension), categorized by highest versus lowest intake and dose–response intake [19–34]. Our systematic search strategy did not yield any studies investigating the potential correlation between SSBs consumption and the conditions of prediabetes and atherogenic dyslipidemia. The age range of the participants in the included systematic reviews was 18–98 years, and both genders were represented in all studies. Across the meta-analyses, the mean minimum and maximum follow-up periods were 1 year and 38 years, respectively. Among the 16 meta-analytic studies included in the analysis, five studies reported a pooled sample size of over 500,000 participants, eight studies featured a minimum follow-up duration of 5 years or greater, while 14 studies reported a maximum follow-up period of 20 years or more. The primary studies encompassed in the meta-analyses were conducted across various regions and were published between 2014 and 2022 (Supplementary Table S3).

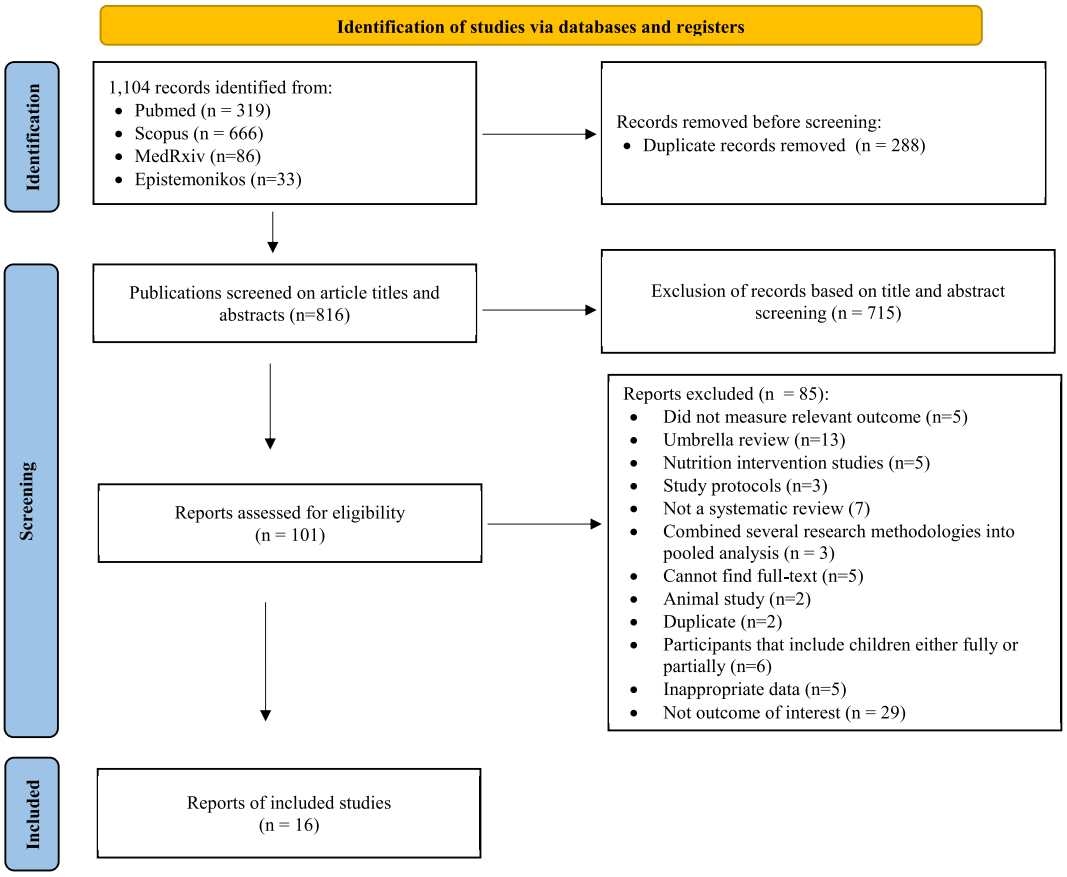


Fig. 1. Flow diagram of included studies according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 guidelines.

3.2. Methodological quality

The results of the methodological quality assessment were presented in Table 1 and Supplementary Figure S1. Of the 16 meta-analyses retrieved, seven were determined to have a 'High' level of methodological quality as evaluated by the AMSTAR 2 tool. The remaining meta-analyses were categorized as 'Moderate' and 'Low' quality levels, with three and six references, respectively. The primary causes leading to downgraded quality ratings were identified as follows [1] failure to specify an established review protocol prior to the study's implementation or insufficient information regarding registered protocols (AMSTAR 2 question 2; 5/16 studies met this criterion) [2], inadequate provision of a list of excluded studies or justification for such exclusions (AMSTAR 2 question 7; 1/16 studies met this criterion), and [3] insufficient reporting of funding sources for included primary studies within most meta-analyses (AMSTAR 2 question 10; 4/16 studies met this criterion).

3.3. Search keywords and study regions

As shown in Fig. 2, relevant keywords related to sugar-sweetened beverages were extracted from the comprehensive appendix outlining the systematic search strategy for the included meta-analysis articles. Moreover, the study locations were also extracted from the detailed description tables of study characteristics in each included meta-analysis. The search terms and geographic locations were visually represented using the "wordcloud" package to illustrate word frequency. In the course of conducting the meta-analysis, the terminology search strategy utilized in relevant databases was centered on specific keywords such as "sweetened", "sugar", and "sugar-sweetened"; "drinks" and "drink"; "beverages", "beverage", and "soft"; "soda", "fruit", "juices", "lemonade", "energy" and "squashes". These keywords were employed either independently or in combination, while utilizing Boolean operators to effectively retrieve relevant primary studies. Additionally, it is worth noting that the study site primarily encompasses developed countries and Europe, such as the United States of America (USA), Sweden, France, Finland, various other European nations and Japan (Supplementary Table S3 and Supplementary Table S5).

3.4. Consuming sugar-sweetened beverages is associated with a heightened risk of MetS in adults

Table 2 presents the reanalyzed estimates with their corresponding 95% confidence intervals (CIs) and the quality of evidence regarding the associations between SSBs consumption and the risk of MetS among adults. Collectively, all evidence strongly supports the association between SSB consumption and increased risk of MetS and its related conditions, with highly suggestive evidence (class II) and convincing evidence (class I).

3.4.1. Obesity

Through a comprehensive and exhaustive search of the literature, we identified three separate systematic reviews that assessed the potential association between SSBs consumption and the incidence of obesity [25,26,28]. These studies provided descriptions of primary studies that employ either or both of the diagnostic criteria

Table 1
AMSTAR 2[a] quality assessment of systematic reviews.

Author and Year	Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	Question 9	Question 10	Question 11	Question 12	Question 13	Question 14	Question 15	Question 16	Overall assessment quality
D. C. Greenwood (2014) [19]	Yes	Partial Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	High
Fumiaki Imamura (2015) [20]	Yes	Partial Yes	Yes	Partial Yes	Yes	Yes	Partial Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
YoungyoKim (2016) [27]	Yes	Partial Yes	Yes	Partial Yes	Yes	Yes	Partial Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Moderate
Pei Qin (2020) [28]	Yes	Partial Yes	Yes	Yes	Yes	Yes	Partial Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	High
Viranda H Jayalath (2015) [29]	Yes	Yes	Yes	Yes	Yes	Yes	Partial Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
Qi Liu (2019) [30]	Yes	Yes	Yes	Yes	Yes	Yes	Partial Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
Bo Xi (2015) [31]	Yes	Partial Yes	Yes	Partial Yes	Yes	Yes	Partial Yes	Partial Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	Low
Lukas Schwingshackl [32]	Yes	Yes	Yes	Partial Yes	Yes	Yes	Partial Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Moderate
Yantong Meng (2021) [33]	Yes	Partial Yes	Yes	Partial Yes	Yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Low
Baoyu Li (2023) [34]	Yes	Yes	Yes	Yes	Yes	Yes	Partial Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	High
Ainara Muñoz-Cabrejas (2023) [21]	Yes	Partial Yes	Yes	Partial Yes	No	Yes	No	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Low
Xiao Zhang (2020) [22]	Yes	Partial Yes	Yes	Partial Yes	No	Yes	Partial Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Moderate
Meng Wang (2014) [23]	Yes	Partial Yes	Yes	Partial Yes	No	Yes	No	Partial Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Low
Vasanti S. Malik (2010) [24]	Yes	Partial Yes	Yes	No	No	No	No	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Low
Leonardo Pozza Santos (2022) [25]	Yes	Yes	Yes	Yes	Yes	Yes	Partial Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	High
D. Ruanpeng (2017) [26]	Yes	Partial Yes	Yes	Partial Yes	Yes	Yes	Partial Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	Low

[a] AMSTAR 2: a critical appraisal tool for systematic reviews that include randomised or non-randomised studies of healthcare interventions, or both [17].



Fig. 2. Search keywords and study regions in included meta-analysis. This figure was produced using Rstudio software.

Table 2
Summary estimates of the associations between SSBs consumption and the risk of MetS, along with 95% confidence intervals and an assessment of the quality of evidence.

Outcomes	Number of meta-analysis studies	Number of primary studies	Number of events	Number of participants	Age range (years)	Follow-up range (years)	Comparison	Recalculated estimates (95% CI[b])	I^2 (%)	Egger's test P value	Quality of evidence
Obesity[a]	3	17	24,750	174,097	18–84	3–20 years	Highest vs lowest	1.18 (1.13–1.24)	0	0.6034	Convincing evidence (class I)
Hypertension[a]	6	45	523,866	1,726,563	18–84	3.24–38 years	Highest vs lowest	1.12 (1.11–1.14)	0	0.7962	Convincing evidence (class I)
Hypertension[a]	6	40	523,257	1,7257,89	18–84	3.24–38 years	Dose–response	1.09 (1.08–1.11)	33.32	0.3097	Convincing evidence (class I)
Type 2 diabetes[a]	6	84	130,505	3,180,169	18–84	3.24–30 years	Highest vs lowest	1.29 (1.25–1.32)	0	0.8722	Convincing evidence (class I)
Type 2 diabetes[a]	6	80	144,199	12,371,582	18–94	3.24–30 years	Dose–response	1.19 (1.16–1.22)	0	0.3721	Convincing evidence (class I)
Metabolic syndrome[a]	3	24	28,221	101,528	18–98	4–10 years	Highest vs lowest	1.29 (1.07–1.52)	75.59	0.1228	Highly suggestive evidence (class II)

[a] Based on meta-analysis of cohorts.
[b] CI confidence interval.

for obesity: a body mass index (BMI) equal to or greater than 30 kg/m^2 and a waist circumference (WC) exceeding 102 cm for men and 88 cm for women. The combined sample size of the reviews included 174,097 individuals, with at least 24,750 reported cases of obesity. The findings from a single study revealed a positive association between the intake of sugar-sweetened beverages and obesity risk. Additionally, a dose–response analysis indicated that an increase in daily SSBs consumption was associated with a higher risk of obesity, with a risk ratio of 1.18 (95% CI: 1.13 to 1.24), representing an 18% increased risk.

3.4.2. Hypertension

A meta-analysis of seven studies was conducted, which demonstrated a consistent association between SSBs consumption and a higher risk of hypertension among a large population of 1,726,563 participants and 523,866 hypertension cases [27–31,34]. Analysis of the highest versus lowest SSBs consumption categories indicated a positive association with hypertension, with a risk ratio of 1.12 (95% CI: 1.11 to 1.14). Moreover, dose–response meta-analysis demonstrated that each increment of one serving/day of SSBs was associated with a 9% higher risk of hypertension in adults, with a risk ratio of 1.09 (95% CI: 1.08 to 1.11).

3.4.3. Type 2 diabetes

We identified six distinct systematic reviews that evaluated the effects of SSBs consumption on the incidence of type 2 diabetes, comparing the highest and lowest levels of consumption [23–25,28,33,34]. The risk of developing type 2 diabetes was observed to be significantly higher in all six meta-analyses, which collectively comprised a study population of 3,180,169 participants and 130,505 instances of type 2 diabetes. Our pooled analysis indicate a statistically significant increase in the risk of type 2 diabetes by 29% (risk ratio 1.29, 95% CI 1.25 to 1.32) in association with the highest versus lowest SSBs consumption. Consistent with the aforementioned findings, our dose–response analysis revealed that an increase in the consumption of SSBs by one serving/day was associated with a statistically significant 19% higher risk of type 2 diabetes (risk ratio 1.19, 95% CI: 1.16 to 1.22) [19,24,28,33–35].

3.4.4. Metabolic syndrome

Three meta-analyses were identified in our comprehensive literature search, reporting the impact of SSBs consumption on the incidence of MetS, which collectively included a total of 101,528 participants and 28,221 cases of MetS [21,22,24]. Our analysis indicates that adults with the highest consumption of SSBs may have an increased risk of MetS by 29% (odds ratio 1.29, 95% confidence

interval: 1.07 to 1.52) compared to those with the lowest consumption.

3.5. Summary effects, heterogeneity and its publication bias

All 30 meta-analytic comparisons investigated in this study were found to be nominally significant at $p < 0.05$. The aforementioned comparisons exhibited varying degrees of heterogeneity, with approximately 60% (n = 18) featuring high heterogeneity (I^2 > 50%), 13.3% (n = 4) displaying moderate heterogeneity (I^2 <50%), 6.7% (n = 2) manifesting small heterogeneity (I^2 <25%), while 20% (n = 6) neglected to report heterogeneity measures. Concerning the matter of publication bias, 43.3% (n = 13) of the included meta-analyses affirmed its absence, with eight studies presenting precise values for the Egger test. Conversely, 23.3% (n = 7) of the studies acknowledged statistically significant publication bias, while 33.3% (n = 10) refrained from acknowledging or discussing it (Supplementary Table S3).

We conducted a reanalysis of heterogeneity in all health outcomes using a random effects model. Our findings indicate no evidence of heterogeneity ($I^2 = 0\%$) in 4 out of 6 outcomes. However, we observed evidence of moderate heterogeneity ($I^2 = 33.32\%$) in the associations between SSBs consumption and hypertension (dose–response), and substantial heterogeneity ($I^2 = 75.59\%$) in the associations between SSBs consumption and MetS (highest vs lowest). Several potential factors could account for the heterogeneity observed across most outcomes, such as differences in settings, regions, ethnicities, sex, ages, study quality, designs, sample sizes, durations of follow-up, and adjustments for confounding factors. Additionally, to determine the presence of publication bias in our reanalysis, we employed Egger's regression test, and found no evidence of bias in any of the health outcomes ($P > 0.1$) (Table 2).

4. Discussion

Our umbrella review presents a comprehensive and up-to-date synthesis of the literature investigating the impact of SSBs consumption on MetS among adult populations. To the best of our knowledge, this study represents the first comprehensive summary and evaluation of the existing meta-analyses on this topic, encompassing methodological quality assessment and an evaluation of the quality of evidence for each association. Through a systematic search strategy, we identified 16 meta-analyses, which collectively generated 30 summary estimates on the associations between SSBs consumption and MetS. The umbrella review yields five key findings. First, both the individual meta-analyses and our reanalysis demonstrate significant associations between SSBs consumption and MetS and its three constituent conditions. Second, we identified five associations that met pre-specified criteria for convincing evidence, while only one meta-analytic estimate met criteria for highly suggestive evidence. Third, there was a lack of consensus on the specific criteria used to identify studies related to SSBs in literature searching. Fourth, the majority of primary studies were conducted in developed countries and Europe. Finally, the methodological quality of most systematic reviews and their meta-analyses for SSBs consumption-MetS associations was high in some cases and low in others.

At present, there is a lack of agreement on a universal definition of SSBs [13], which is also evident in the search terms used in the meta-analyses included in our review. The investigation of the association between SSBs consumption and health outcomes was the primary aim of the meta-analyses included in our review. However, discrepancies in definitions and inconsistent utilization of search terms may have arisen. For instance, one study exclusively employ the term “sugar-sweetened beverages” without indicating the specific type of beverage [32], while others employ only a few related terms [20,29,33]. Conversely, certain studies use a combination of various terms related to SSBs to identify relevant literature [25,31,36]. Upon comparing the search keywords employed in the included meta-analyses with the definition of SSBs beverages provided by the WHO [11], it was observed that some meta-analyses may have insufficiently included related terms, while others may have included an excess. Moreover, this observation underscores the wide range of SSBs products that are commercially available, thus necessitating the development of more precise definitions based on sugar content per volume to better evaluate the causal association between SSBs consumption and health outcomes, rather than relying solely on generic labeling of different types of SSBs and its volumes. However, even if the available evidence has focused on examining the association between individual types of SSBs and MetS, it is important to acknowledge that in reality, people consume a combination of beverages and nutrients that may interact with each other. Furthermore, MetS is a complex and multifactorial disorder that arises from a high caloric intake, sedentary lifestyle, and central obesity [3]. While identifying the potential detrimental effects of individual SSBs and their dose responses are useful in elucidating the underlying biological mechanisms, it is important to recognize that no single beverage or nutrient is solely responsible for the overall effect. Therefore, the findings of our review, which provide a robust synthesis of the available evidence on the association between SSBs consumption and the risk of MetS, should be taken into account in explaining this relationship.

Our location analysis results are in accordance with the epidemiological data on the worldwide prevalence of SSBs consumption. Classical examples of the countries analyzed, the United States of America (USA) was included in all of the meta-analyses. Based on national survey data from the USA, it appears that SSBs intake has experienced a modest decline since the early 2000s [37], though consumption levels remain alarmingly high, with 63% of adults aged 18 or older reporting daily or more frequent consumption of SSBs [38]. Globally, regions such as North America, Latin America, Australasia, and Western Europe exhibit the highest levels of SSBs [12]. Consistently, our finding also indicate that the investigation of the association between SSBs consumption and health outcomes is predominantly conducted in developed countries, while limited evidence exists in LMICs. Notably, this issue warrants greater attention as recent studies have shown that SSBs consumption is generally higher in LMICs. Specifically, an analysis of available data found that SSBs consumption was higher in upper-middle income and lower-middle income countries compared to high income or low income countries [39]. This phenomenon may be attributed to the rapid urbanization and economic development in many LMICs, which has led to increased accessibility and availability of SSBs [40].

Drawing from available evidence from long-term prospective cohort studies assessing outcomes, our synthesis indicates robust evidence for an etiological between the consumption of SSBs and the increased risk of obesity, type 2 diabetes, hypertension, and MetS in adult populations. First, based on the congruence of our findings, an analysis of historical data from the USA demonstrates parallel trends between the increase in consumption of added sugars (predominantly from SSBs) and the epidemics of obesity and type 2 diabetes [41]. Additionally, the decrease in SSBs consumption since the early 2000s has not resulted in a reduction in the prevalence of obesity. However, this observation should not be considered as evidence refuting the established association between SSBs consumption and weight gain. Second, a significant amount of scientific literature has demonstrated a positive association between the consumption of SSBs and an increased risk of type 2 diabetes, attributable to both weight gain and other

independent metabolic pathways. When estimates that did not adjust for BMI were used in the analysis, a meta-analysis of 17 prospective cohort studies found that an increase in SSBs intake of one serving per day was associated with an 18% higher risk of type 2 diabetes (95% CI 9–28%). However, this association was attenuated to 13% (95% CI 6–21%) when estimates that adjusted for BMI were included, suggesting that BMI partially mediates the association [35]. These findings, along with our findings and more recent studies published, strongly support the association between SSBs consumption and increased risk of type 2 diabetes in diverse populations, such as Asia [42], USA and Europe [23], and sub-Saharan Africa [43]. Third, an important inquiry arising from our findings is whether the elevated risk of hypertension is a direct consequence of consuming SSBs, or if the caloric content of SSBs contributes to weight gain, consequently augmenting the risk of hypertension indirectly. To elucidate this point, both our study and previous epidemiology cohort studies concur that the consumption of SSBs is an independent or/and co-risk factor for hypertension. In the study conducted by Kristin Øksendal Børresen, upon excluding individuals with the highest weight gain at 6 months postpartum, a stronger association between SSBs intake and subsequent hypertension was also observed. Conversely, after adjusting for potential confounding variables, a decrease in SSBs intake of one serving per day was associated with a reduction of 1.8 mm Hg (95% CI: 1.2 to 2.4) in systolic blood pressure and 1.1 mm Hg (95% CI: 0.7 to 1.4) in diastolic blood pressure over a period of 18 months [44]. Notably, this association was also partially mediated by changes in weight and the underlying mechanism through which an increased consumption of SSBs may elevate blood pressure remains uncertain [44]. Fourth, our findings align with previous studies that have examined the association between SSBs consumption and MetS, despite the existence of methodological discrepancies. For example, a study that assessed the metabolic health of 5107 participants from the Framingham Heart Study Offspring and third-generation cohorts found that regardless of weight status, SSBs consumers had a greater likelihood of developing metabolic disorders such as hypertension, insulin resistance, high fasting glucose and triglycerides, and lower high-density lipoprotein cholesterol levels [45]. It is noteworthy that we were unable to compare the number of servings due to the lack of comparable data across studies, and therefore only extreme categories of SSBs consumption were analyzed. This may be attributed to a limited number of prospective studies that have investigated the association between SSBs consumption and the development of MetS, which can be partly explained by the challenges encountered in accurately assessing the outcome. Finally, although common threshold for intake of SSBs has not been determined from dose–response analyses in included meta-analyses, it is evident that even at commonly consumed levels, such as one serving per day or 250 mL/day, intake of SSBs is associated with clinically significant weight gain and increased risk of MetS and its related conditions.

4.1. Strengths and limitations

Our umbrella review demonstrated several noteworthy features. First, it presented a comprehensive synthesis of meta-analyses investigating the association between SSBs consumption and MetS. Second, we employed validated tools to assess the methodological quality for all identified associations.

However, our study also has some limitations. Firstly, our review acknowledged the broad and heterogeneous nature of the evidence synthesized. Although heterogeneity is a relevant indicator of potential bias in this literature, it can also represent genuine heterogeneity within and between MetS conditions. Furthermore, discrepancies in the search keywords employed by individual meta-analysis studies may contribute to the observed heterogeneity. Secondly, we relied on the information presented in the included meta-analyses and utilized the AMSTAR 2 tool to assess the quality of evidence for each association. This approach inherited the limitations of the included studies, such as the presence of serious or very serious risk of bias that could downgrade the quality of evidence. Consequently, our umbrella review may be limited in its ability to furnish robust evidence for the development of preventive recommendations against MetS by restricting SSBs intake. Thirdly, due to access limitations, we were unable to explore additional databases such as CINHAL, Embase, and MEDLINE. Consequently, there is a potential for bias due to our restricted search approach. Finally, due to the unavailability of raw data, we were unable to identify any meta-analytic studies exploring the association between SSBs consumption and other conditions of MetS, namely prediabetes and atherogenic dyslipidemia.

5. Conclusion

Our umbrella review systematically evaluated the available systematic reviews and meta-analyses to comprehensively assess the adverse consequences of SSBs consumption on MetS and its components among the adult population worldwide. This findings are useful for quantifying disease burden related to beverage intake and shaping health policies and interventions aimed at mitigating relevant diseases risks. In the upcoming years, as the world confronts escalating burdens of obesity and chronic disease alongside infectious diseases, there will be an immediate need for coordinated actions across all sectors of society to prioritize prevention of chronic diseases such as MetS. Future research should focus on examining the effects of various SSBs on health outcomes across a wide range of doses, particularly in LMICs. Considering the robustness and consistency of the evidence across diverse populations, coupled with the observed rise in consumption associated with nutrition transitions, SSBs represent an unequivocal target for policy intervention. Hence, the efforts should concentrate on nutrition policies and regulatory strategies aimed at enhancing overall diet quality, creating healthier food environments, and reducing health disparities between nations and regions. Given the widespread consumption of SSBs globally, reducing the intake of beverages constitutes a crucial step towards improving dietary quality, having a significant impact on weight management, mitigating the risk of MetS, and ultimately enhancing global health outcomes.

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Author contributions

- TQD, LCL and PNQ: Conceptualization, Methodology.
- TQD, HLV, BTT and LCL: Data curation, Writing- Original draft preparation.
- PNQ, VTQC and NTHH: Visualization, Investigation.
- PNQ, HLV and TQD: Supervision.
- VTQC, LCL and NTTH: Software, Validation.
- All of the authors: Writing- Reviewing and Editing.

Ethics approval

Since this was a review study, no permission from the ethical committee was necessary.

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clnesp.2023.08.001.

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